Halil İbrahim Ergül

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Experience

OLX May 2025 – Present
Senior Data Scientist *Netherlands, Amsterdam (Remote)*

- Boosted purchase conversions by 1% and reduced zero-result problem by 3% by deploying an LLM-powered query agent across 3M+ daily searches.
- Delivered 4% growth in purchase and click conversions by rebuilding a multi-label classification pipeline (via AWS Batch, Airflow, MLflow) to land users into more relevant product categories across 7 global markets.
- Increased online model performance by curating high-quality labeled datasets for training and evaluation, iterative model training, and production deployment.
- Conducted exploratory analysis and statistical evaluation to produce stakeholder reports that directly shaped roadmap decisions.

Hepsiburada (NASDAQ: HEPS) May 2024 – May 2025
Data Scientist II *Türkiye, Istanbul*

- Developed N-gram and topic extraction-based search box suggestions, driving a 10% lift in click-through and 7% higher order conversions on a platform serving millions of daily active users.
- Improved search relevance through point-wise ranking experiments: engineered 50+ features, deployed updated models via Docker, that resulted in 2% online metric gains.
- Debugged NER model that achieved 6% greater entity recognition accuracy to improve query understanding quality.
- Built systematic evaluation pipeline for typo correction model using LLM-based annotation
- Designed and launched 'Related Search Term' feature (weekly-refreshed, serving all users post-A/B) using BERT and recommendation algorithms to surface relevant keywords on listing pages and increasing purchase conversions by 2%.

Behavioral Analytics and Visualization Lab 2022 – 2024
Research Assistant (M.Sc.) *Türkiye, Istanbul*

- Fine-tuned open source LLMs for customer behavior classification and synthetic data generation.
- Worked on an NLP project for Turkish sentiment tone detection with BERT models, supported by TUBITAK.

Upily Jan 2021 – May 2024
Data Scientist II & ML Engineer *Türkiye, Istanbul*

- Built automated and scheduled ETL pipelines for processing millions of daily events via BigQuery and creating dashboards.
- Ad-hoc data analysis, user clustering, and developed end-to-end ML models for churn prediction to improve conversion rates.
- Collaborated with Product, Marketing, and Growth teams on ad-hoc A/B test results analysis and custom KPI reporting for 5+ monthly experiments.
- Scraped and processed 500+ hours of audio to train in-house TTS voices to accelerate art department content creation.
- Engineered and deployed an on-premises LLM assistant to automate internal workflows and to accelerate the enrichment of app content for product teams.

Education

Sabancı University Graduation Date: 2024
M.Sc with Thesis, Data Science, GPA: 3.60/4.00 *Türkiye, Istanbul*

Thesis Title: Instruction-based fine-tuning of open-source LLMs for predicting customer purchase behaviors

Boğaziçi University Graduation Date: 2021
BA, International Relations & Sociology, GPA: High Honors, 3.94/4.00 (1st among cohorts) *Türkiye, Istanbul*

Skills

ML/DL: Python, PyTorch, Scikit-learn, Transformers, MLflow
Data Engineering: SQL, PySpark, Airflow, BigQuery, ElasticSearch, Redis
Cloud: AWS (S3, Batch, Athena), GCP
Tools: Docker, Prometheus, New Relic, Tableau, Looker, GA4

Projects

RAG-based Question-Answer System | *LLamaIndex, ElasticSearch, FastAPI, Docker*

- Built a production-ready RAG pipeline that ingests PDF documents, applies semantic chunking, indexes embeddings in ElasticSearch, and serves answers with source citations via FastAPI and containerized with Docker for easy deployment.

LLM Fine-Tuning with PEFT | *LLaMA3, LoRA, SFT, Hugging Face*

- Fine-tuned LLaMA3-instruct on a custom dataset using LoRA and supervised fine-tuning (SFT), with detailed guidance on data formatting, EOS token handling, and LoRA hyperparameter tuning for stable training.

NLP-Powered Hotel Recommendation System | *BERTopic, Turkish NLP, Cosine Similarity*

- Built a two-stage recommendation engine: retrieves top-10 hotels via cosine similarity on BERTopic embeddings of Turkish descriptions, then re-ranks using a composite popularity score achieving 75% geographically relevant recommendations.

Publications

The Impact of RFF and Ensemble Modeling Approach in an E-commerce Product Ranking, IEEE

Instruction-Based Fine-tuning of Open-Source LLMs for Predicting Customer Purchase Behaviors, arXiv

LDA Topic Modeling on Twitter Data Concerning Immigrants and Refugees, IEEE

Certificates

Advanced Learning Algorithms, AWS Cloud, ML in Production, Big Data & Business Analytics, NN and Deep Learning